

NORTHERN TERRITORY SALTWATER CROCODILE (Crocodylus porosus) RISK MANAGEMENT FRAMEWORK 2021 26

Department of
ENVIRONMENT, PARKS AND WATER SECURITY

PARKS AND WILDLIFE COMMISSION NORTHERN TERRITORY

NORTHERN TERRITORY GOVERNMENT

MANAGEMENT ACTIONS REDUCE THE RISK OF CROCODILE ATTACK BUT CAN NEVER GUARANTEE SAFETY.

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Enquiries should be made to:

Department of Environment, Parks and Water Security
Email: Communications.DEPWS@nt.gov.au
Web: depws.nt.gov.au
Mail: PO Box 496, Palmerston NT 0831

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INTRODUCTION

Saltwater crocodiles live in the Top End of Northern Australia. Any body of water in the Top End has the potential to contain large and dangerous saltwater crocodiles. The common name 'saltwater crocodile' has created misconceptions that they only live in saltwater, however they can thrive in both salt and fresh water and can be found in freshwater hundreds of kilometres inland.

The Parks and Wildlife Commission (Parks and Wildlife) provides a program of crocodile management and public education that is designed to reduce the risk of human/crocodile interactions that will result in the injury to or fatality of people and domestic animals.

The management program includes not only those areas set aside for visitor use in National Parks and Reserves but also the broader environment of the crocodile habitat area across the Northern Territory (NT) including Darwin Harbour and the Katherine River.

The objectives of the saltwater crocodile management program are as follows:

- Reduce the likelihood of crocodile attack.
- Warn the community of potential presence of crocodiles
- Educate the community about safe behaviour around waterways
- Reduce the number of saltwater crocodiles entering areas where they could come into contact with people.

Crocodiles are culturally and economically important to Aboriginal people, contribute to a healthy natural environment, and play a significant part in the Territory's economy through tourism, crocodile farming and wild harvesting.

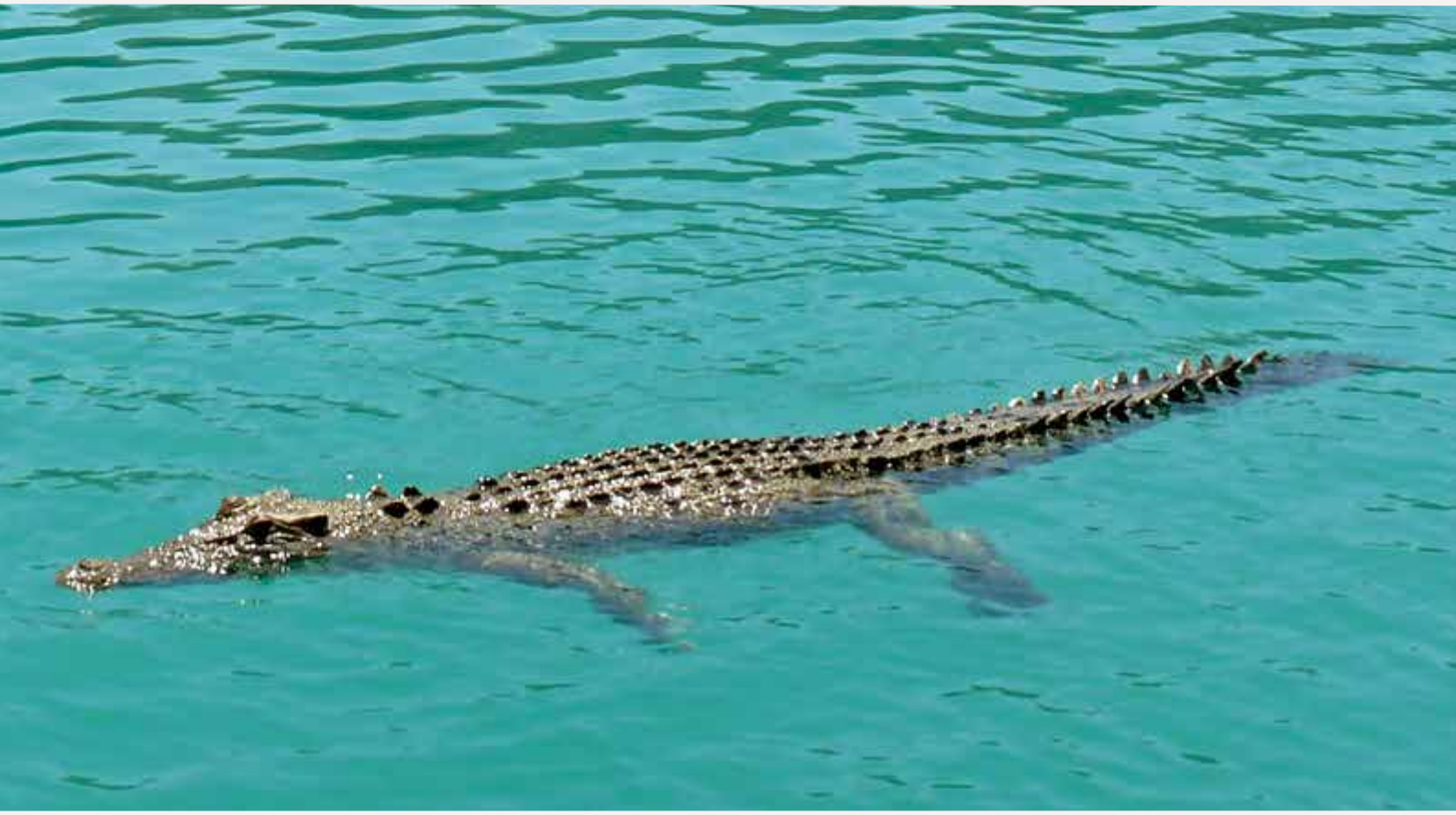


IMAGE: Saltwater crocodile at Prince Fred Harbour. Photo by Ian Morris

PURPOSE

This document outlines the risk management framework that Parks and Wildlife use to guide crocodile management and public education aimed at reducing problem human/crocodile interactions.

	GENERAL MANAGEMENT	BARRIER AND REMOVAL	ACTIVE REMOVAL
RISK ASSESMENT			
ZONES	Areas that are typical habitat for crocodiles but are not near a large urban centre. Public education is carried out and problem crocodiles are removed as required.	Applied to areas where recreational water activities are supported and there are natural/artificial physical barriers that are generally effective in preventing crocodiles from entering an area.	Used in the broader landscape of the NT where there are high human populations in proximity to areas of rivers, harbours, creeks and wetlands where crocodiles are frequently found.
PLANNING			
ACTIONS	Public education and awareness campaigns in combination with targeted removal of problem crocodiles.	Individual management plans that consist of maintenance and monitoring of natural/artificial barriers, night surveys, trapping and public communications and interpretive signage.	Significantly reduce the number of crocodiles in close proximity to large urban areas regardless of size or behaviour with active stakeholder engagement and partnerships.



ASSESSING RISK

Risk is assessed using the following factors:

- the size of the human population in an area
- the size of the crocodile population in an area
- whether crocodiles are resident or transitory in an area
- the viability of various management options in an area.

A visual representation of how risk is assessed is provided in Figure 1 below.

The criteria for determining if there is a significant level of risk that requires a management intervention in an area of the NT is established by the overall risk to public safety present there.

For example, with the increase in population in an area, the probability of a crocodile attack increases in line with the number of people using the waterways even without an increase in crocodiles. An increase in human population associated with known proximity to a permanent and breeding crocodile habitat maximises the risk of a problem human/crocodile interaction.

In response to the varying levels of risk that can be found across the northern NT due to the different population densities of people and crocodiles, the different way people use their waterways, and the public expectations of safety, Parks and Wildlife have developed three distinct crocodile management zones with specific objectives for each zone.

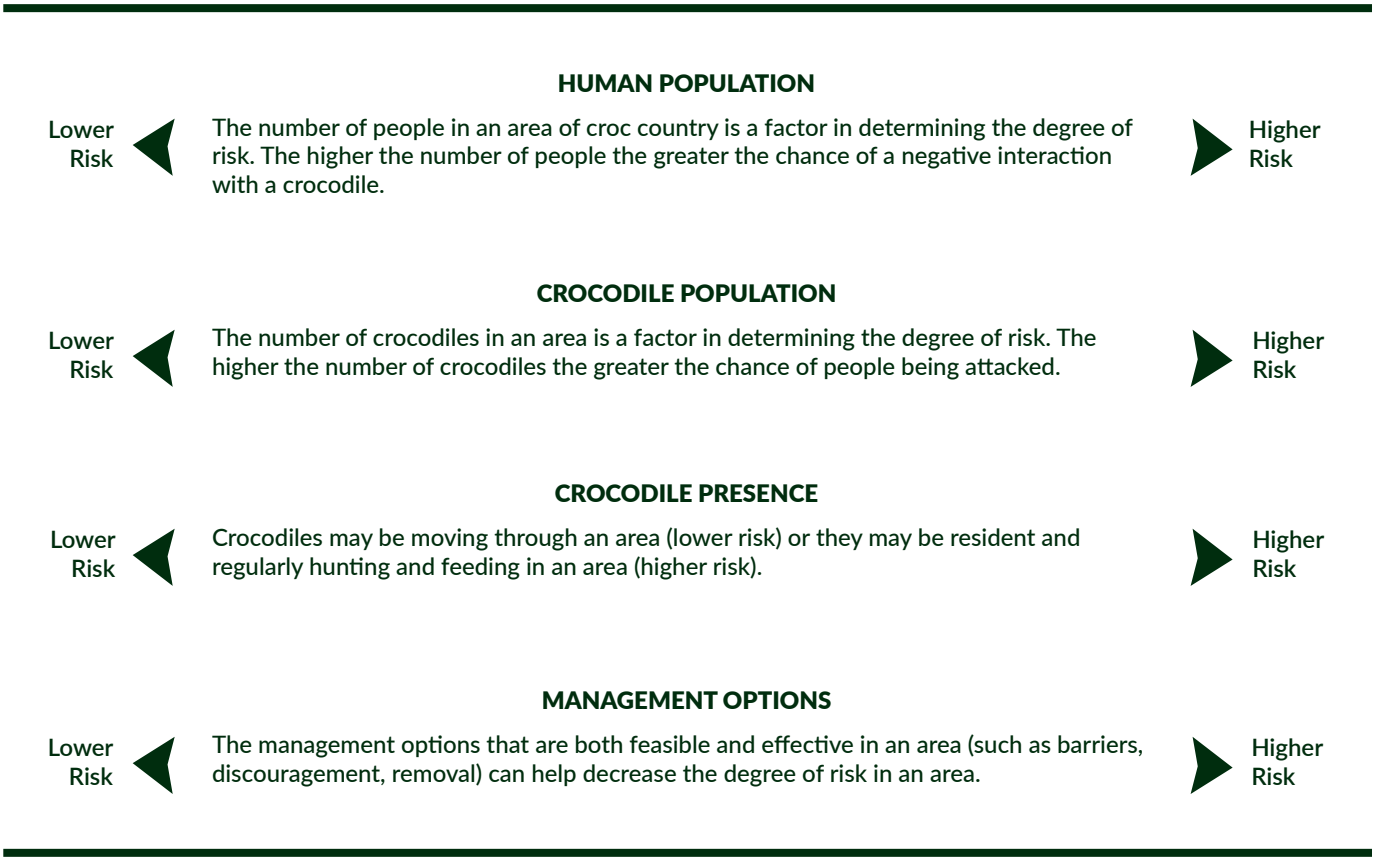


Figure 1: Risk Determination Tool (Key factors in determining risks to public safety, reproduced from the Queensland Crocodile Management Plan 2017, Department of Environment and Science)

ZONES

A ‘general management zone’ is used in those areas that are typical habitat for crocodiles but are not near a large urban centre. Public education is carried out and problem crocodiles are removed as required.

An ‘active removal zone’ is used where there are high human populations in proximity to rivers, harbours, creeks and wetlands where crocodiles are frequently found. An urban centre close to bodies of water encourages water-based recreational activities and management of both human behaviours and crocodile numbers is implemented. The objective in this zone is to generate the highest possible levels of safe behaviours, while also removing a number of crocodiles from the environment, therefore lessening the risk. It should be noted that the risk profile in these areas cannot be brought lower than a medium risk of a problem human/crocodile interaction as there will always remain the possibility that a crocodile is in the environment.

A ‘barrier and removal zone’ is suited to areas where there are physical barriers that are generally effective in preventing crocodiles from entering an area. The objective in this zone is to use natural barriers and/or install artificial barriers where it is feasible to do so, such that swimming holes and pools can be surveyed and declared as ‘low risk’ for water recreation based on negative survey results. These zones are only applied to pools and waterholes on lands managed by Parks and Wildlife due to the high level of infrastructure and resourcing required to manage crocodile risk in this way.



The General Management Zone

This zone includes all land, townships and communities across the Top End of the NT where saltwater crocodiles may be present. A map of the area is provided in Map 1: Northern Territory General Management Zone for Crocodile Management.

Parks Australia is responsible for the management of saltwater crocodiles in Kakadu National Park and also provide support to the township of Jabiru.

Risks posed by crocodiles

The risk of human/crocodile interaction in this zone is variable due to the range of habitats and varying numbers of people found across the landscape. Large crocodiles or crocodiles displaying dangerous behaviour, are often living in this zone and the population is completely unmanaged apart from removing problem crocodiles that come to the attention of authorities. The overall risk is reduced by the low levels of human population in many areas that have high crocodile densities.



Management response

The crocodile management response in this zone relies primarily on public education in combination with targeted removal of problem crocodiles.

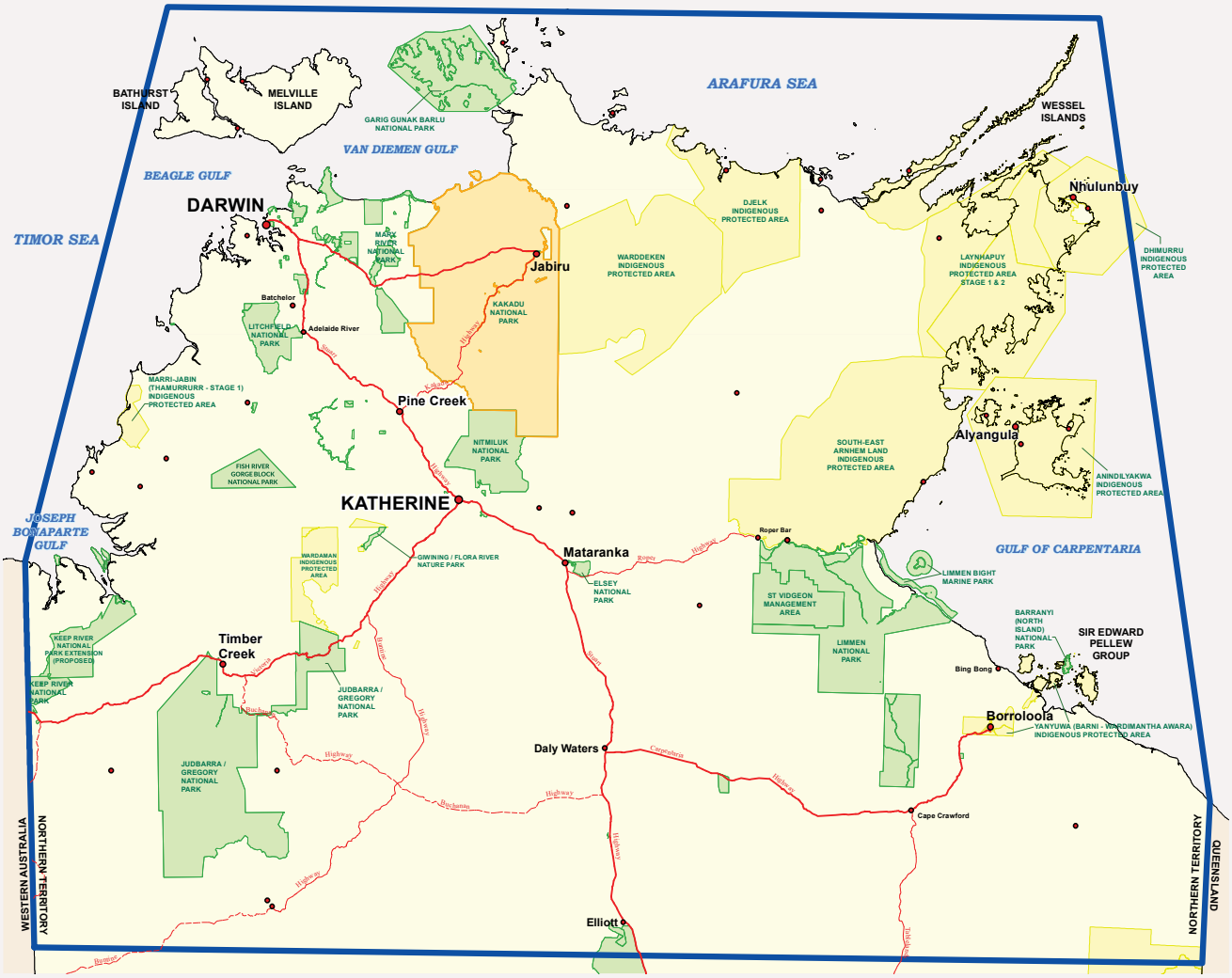
A problem crocodile is one that:

- has attacked or has been reported for attempting to attack a person
- behaves aggressively towards a person
- is at a site where it is a potential threat to human safety or wellbeing or
- is affecting the productivity of industry or commercial enterprises.

While the removal of problem crocodiles from this zone reduces the likelihood of attacks from individual crocodiles, the chance of a crocodile attack remains high, particularly in areas of high quality crocodile habitat. People living in or visiting areas in this zone need to stringently practise Be Crocwise behaviour to reduce the likelihood of a crocodile attack.

Zone Map

The map below shows the Northern Territory General Management Zone for Crocodile Management.



Map 1: Northern Territory General Management Zone for Crocodile Management

LEGEND

- NT General Management Zone
- Park or Reserve
- Australian Government and Traditional Owners Managed Park
- Indigenous Protected Area

The Active Removal Zones

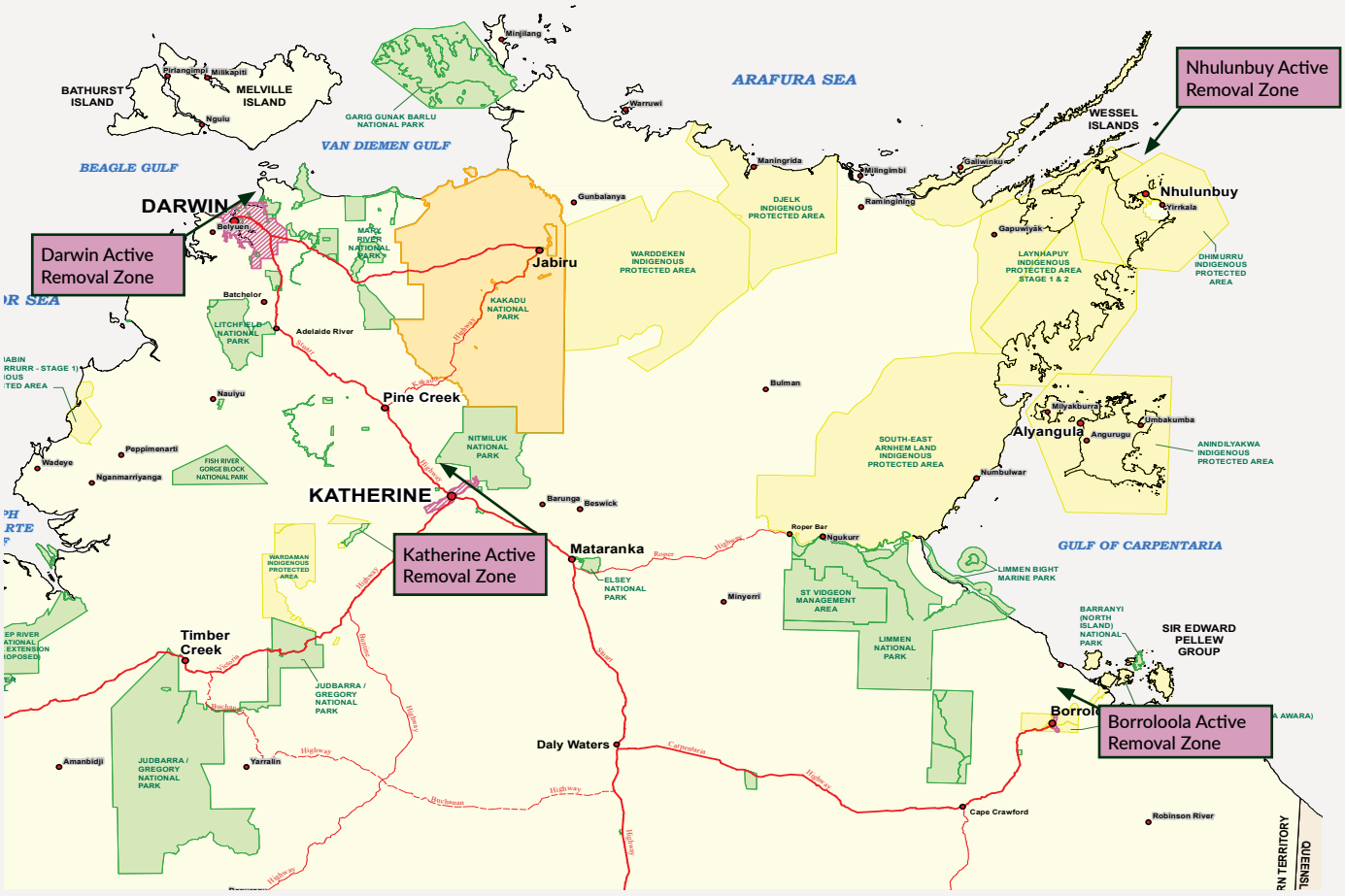
The Active Removal Zone consists of four areas as can be seen on Maps 2 and 3: Northern Territory Active Removal Zones for Crocodile Management.

- Darwin Active Removal Zone - Darwin harbour, Darwin city, Palmerston and the rural area.
- Katherine Active Removal Zone – Along the Katherine River from the Nitmiluk National Park boundary past the Katherine Township, to the junction of the Katherine River and Vampire Creek.
- Borroloola Active Removal Zone - A 13 kilometre section of the McArthur River adjacent the Borroloola Township.
- Nhulunbuy Active Removal Zone – The town lagoon area and sewerage ponds.

The Darwin, Katherine and Borroloola Active Removal Zones are managed by Parks and Wildlife while the Nhulunbuy Active Removal Zone is managed by the Nhulunbuy Corporation. Each of these four zones have crocodile management risk plans.

Risks posed by crocodiles

The risk of human/crocodile interaction in this zone is high. This is due to the significant number of people living in the area and the high likelihood of crocodiles entering the area from surrounding crocodile habitats. There is a constant pressure for increased opportunities around water based recreation, particularly in Darwin harbour where people are known to paddle, swim, surf, kayak, dive, jet ski and boat in the area. These practices, whilst not sanctioned by authorities managing waterways and crocodiles, serve to increase the risk markedly so that a significant management response has been implemented.



Map 2: Northern Territory Active Removal Zones for Crocodile Management

LEGEND

- NT Active Removal Zones
- Park or Reserve
- Australian Government and Traditional Owners Managed Park
- Indigenous Protected Area

Management response

The management objective for this zone is to significantly reduce the number of crocodiles in close proximity to large urban areas. This is done by implementing a schedule of routine trapping and removing animals, patrolling to detect and capture crocodiles, and opportunistically catching and removing them when reported in the area. All crocodiles that are trapped or captured are removed, regardless of size or behaviour.

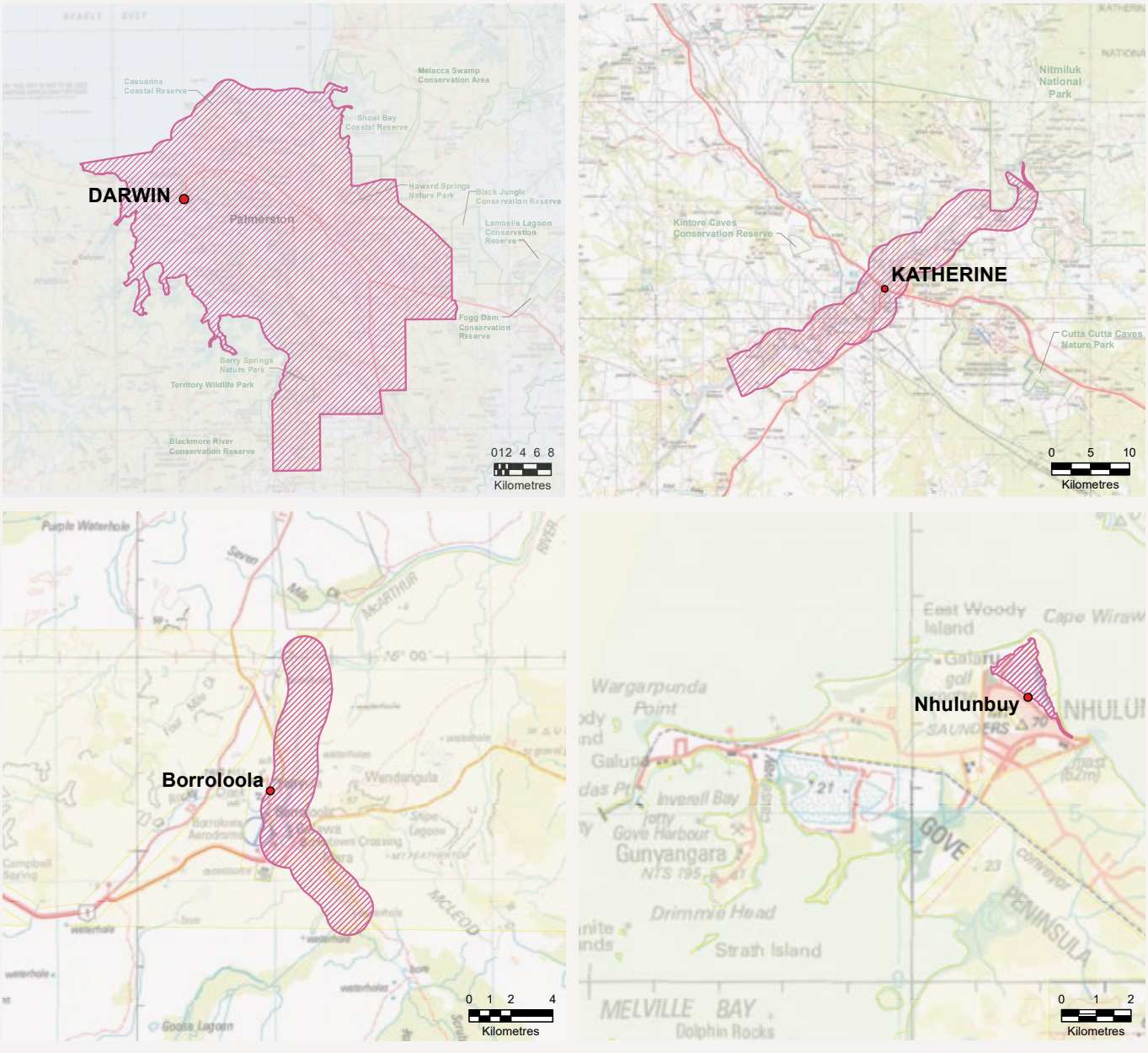
These actions mitigate the risk to some extent however public education remains a key priority. At best, after a number of crocodiles are removed from these zones, the risk can only be reduced to medium. Crocodiles are wild animals that can be difficult to detect and capture, and can swim long distances.

At any given time, there may be undetected crocodiles already within the zone, known crocodiles yet to be removed, and new crocodiles entering the zone.

Even when a crocodile has been removed, it leaves a vacated territory for another crocodile to fill. People living in or visiting areas in this zone need to practice Be Crocwise behaviour to reduce the likelihood of a crocodile attack.

Zone Map

The map 2 shows the four Active Removal Zones with map 3 showing a inset view of these zones.



Map 3: Insets from map 2 Northern Territory Active Removal Zones for Crocodile Management

The Barrier and Removal Zones

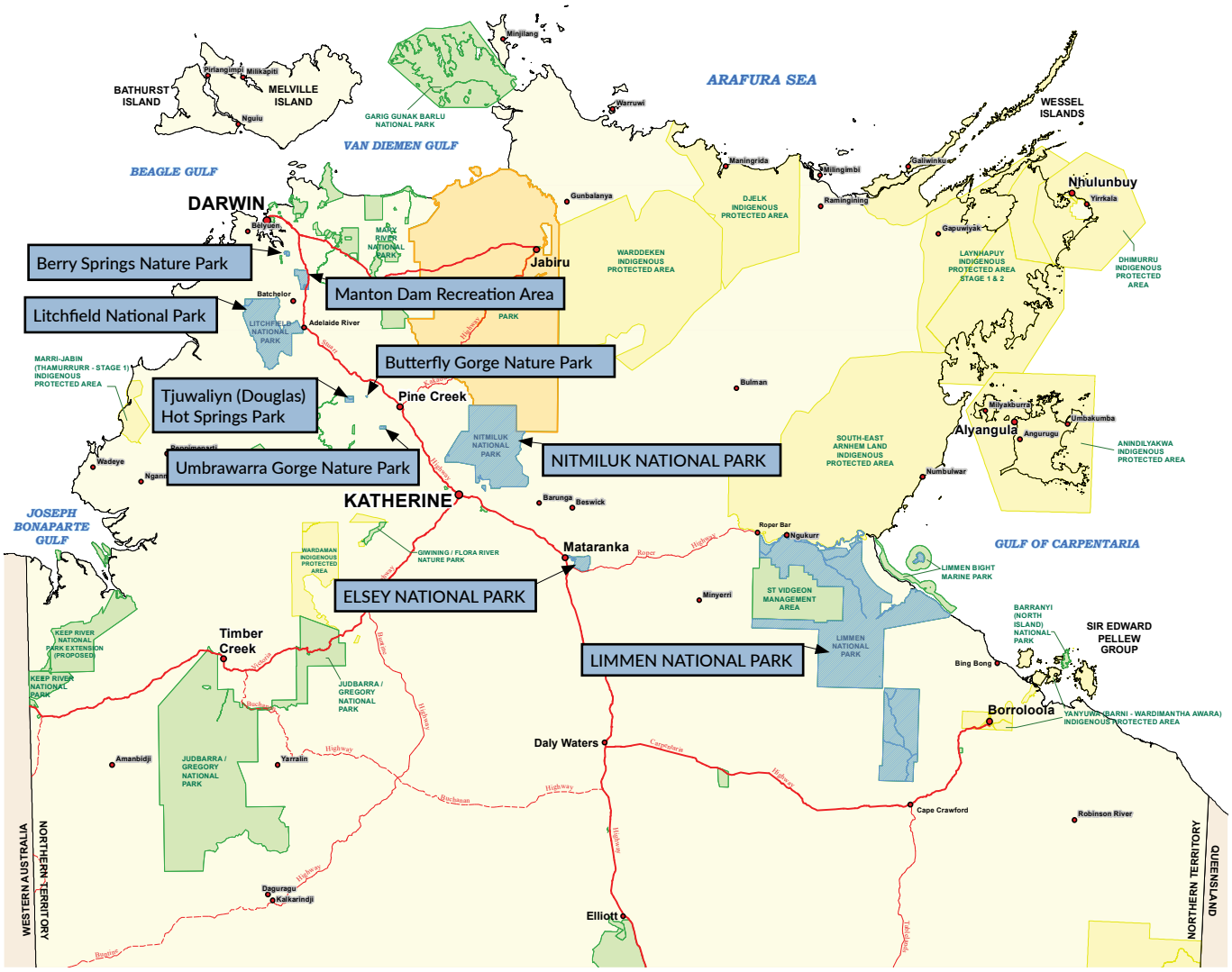
The Barrier and Removal Zones comprise 9 parks and reserves where recreational water activities are supported (refer Map 4: Northern Territory Barrier and Removal Zones for Crocodile Management). Each park or reserve under this management zone has an individual crocodile management risk plan.

Parks that have swimming sites and areas managed as barrier and removal zones are as follows:

- Berry Springs Nature Reserve
- Butterfly Gorge Nature Park
- Elsey National Park
- Litchfield National Park
- Limmen National Park
- Manton Dam Recreation Area
- Nitmiluk (Katherine Gorge) National Park
- Tjuwaliyn (Douglas) Hot Springs Nature Park
- Umbrawarra Gorge Nature Park.

Zone Map

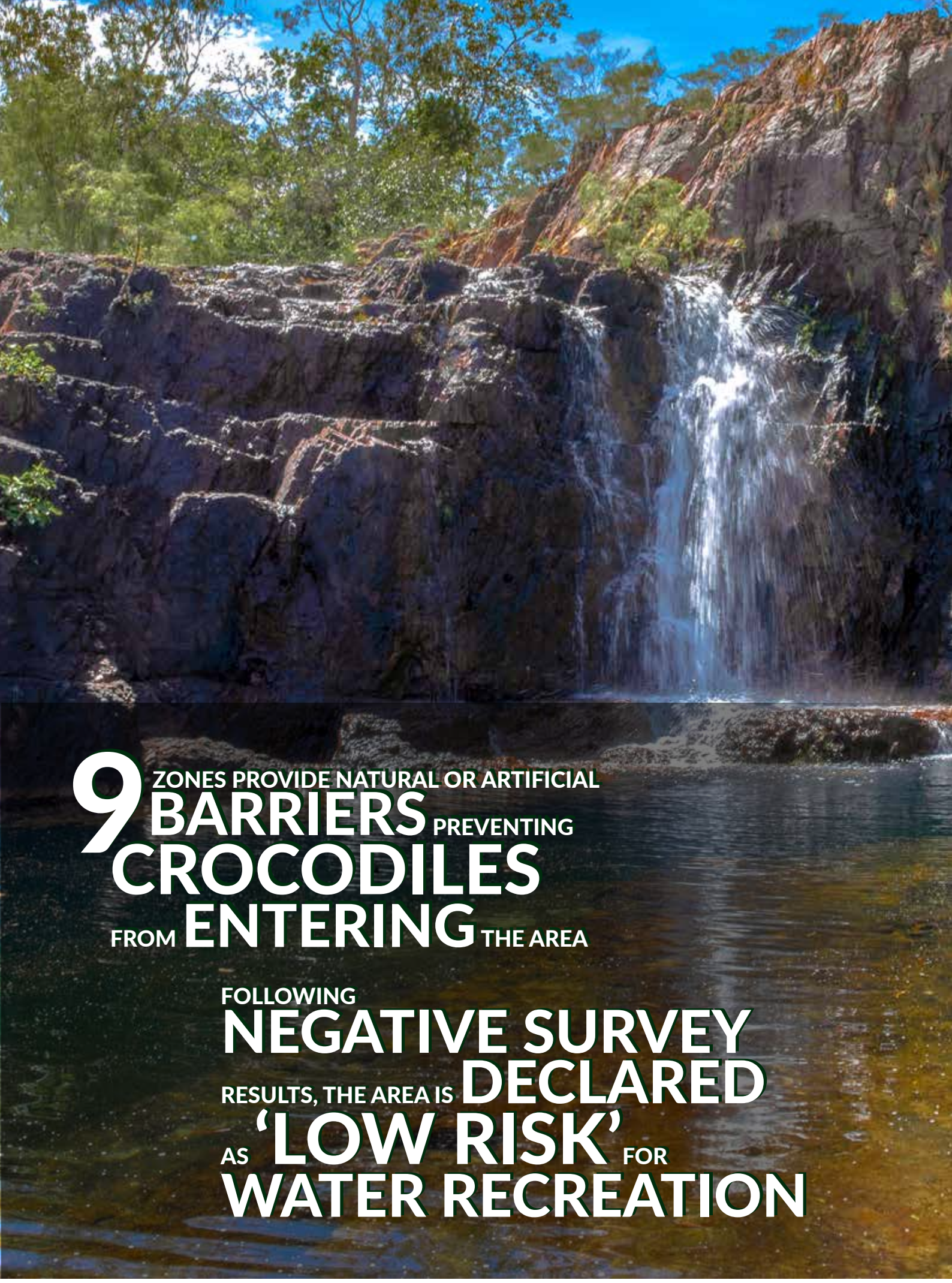
The map below shows the nine Northern Territory Barrier and Removal Zones for Crocodile Management.



Map 4: Northern Territory Barrier and Removal Zones for Crocodile Management

LEGEND

- NT Barrier and Removal Zone
- Park or Reserve
- Australian Government and Traditional Owners Managed Park
- Indigenous Protected Area



9 ZONES PROVIDE NATURAL OR ARTIFICIAL **BARRIERS** PREVENTING **CROCODILES** FROM **ENTERING** THE AREA

FOLLOWING **NEGATIVE SURVEY** RESULTS, THE AREA IS **DECLARED** AS **'LOW RISK'** FOR **WATER RECREATION**

CROCODILE MANAGEMENT ACTIONS

Saltwater crocodiles detected in the Darwin, Katherine and Borroloola Active Removal Zones are removed by the Crocodile Management Unit, Wildlife Rangers or Rangers and recorded in the Parks and Wildlife Crocodile Capture database (see Table 2). Saltwater crocodiles captured in the Nhulunbuy Active Removal Zone are removed by the Nhulunbuy Corporation. In the General Management Zone only problem crocodiles are removed. This action may be undertaken by the Crocodile Management Unit, Wildlife Rangers, local councils, Aboriginal Ranger Groups, Aboriginal Custodians, permitted contractors or private land holders.

Parks Australia is responsible for the management of saltwater crocodiles in Kakadu National Park and also provide support to the township of Jabiru.

Table 2: Crocodiles caught in the Active Removal Zones over the last 9 years (Crocodile Capture Database).

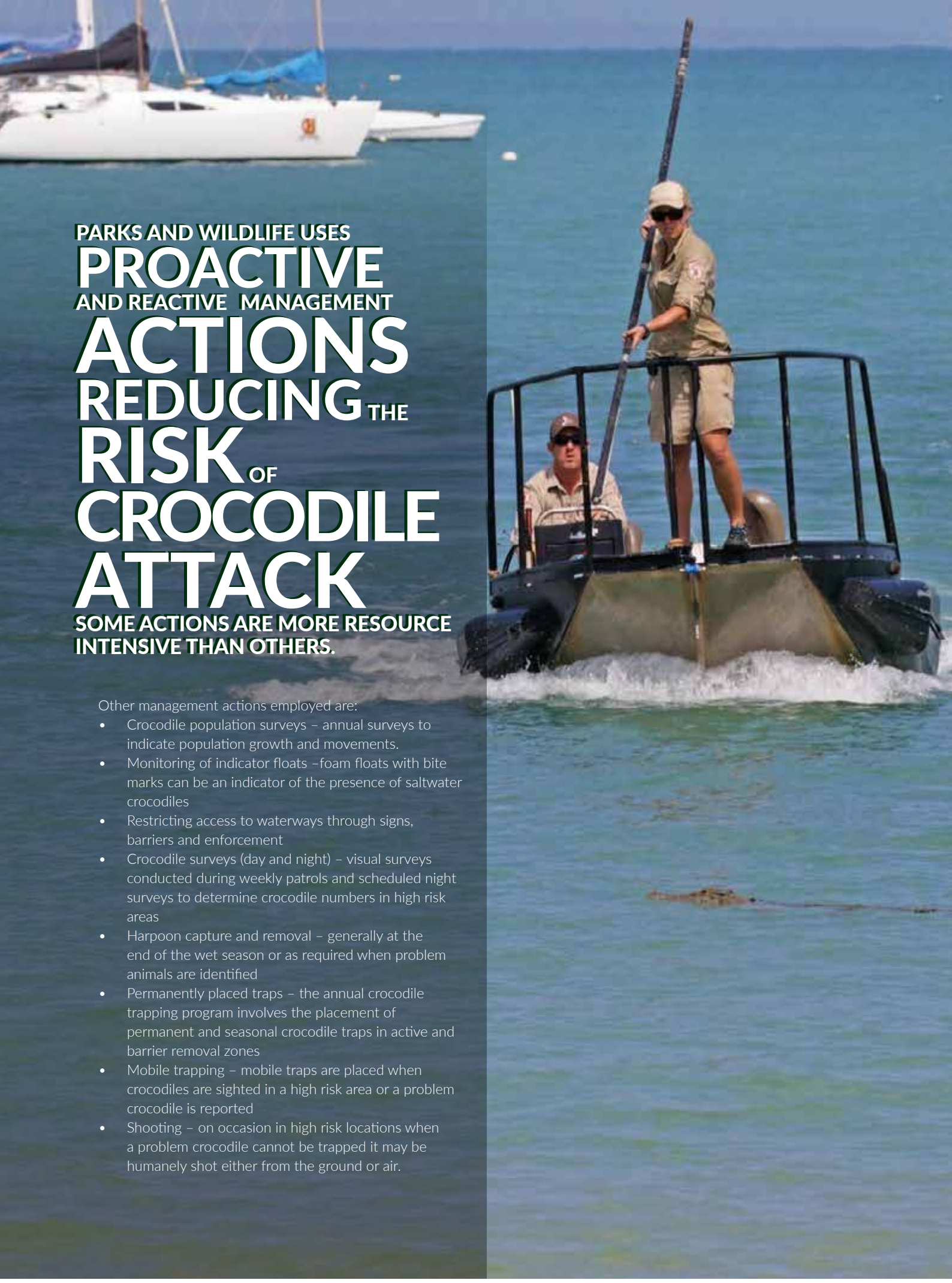
Year	Darwin Active Removal Zone			Katherine Active Removal Zone			Borroloola Active Removal Zone			Nhulunbuy Active Removal Zone		
	Crocs caught	Av Size cm	Largest size cm	Crocs caught	Av Size cm	Largest size cm	Crocs caught	Av Size cm	Largest size cm	Crocs caught	Av Size cm	Largest size cm
2011	244	232	451	4	262	464	2	256	274	-	-	-
2012	273	227	481	2	372	418	0	0	0	-	-	-
2013	186	212	425	4	282	305	6	295	454	-	-	-
2014	248	235	425	1	391	391	1	173	173	-	-	-
2015	251	220	410	1	353	353	19	290	393	-	-	-
2016	223	202	400	3	304	331	3	378	460	-	-	-
2017	349	221	418	4	302	387	4	232	396	-	-	-
2018	330	219	472	6	358	422	5	336	472	11	194	228
2019	211	210	426	0	0	0	6	316	457	18	216	351
Total	**2315**			**25**			**46**			**29**		

Note: The figures in Table 2 are as of 14 January 2020. Data from Parks and Wildlife Crocodile Capture Database.

Management Actions

Parks and Wildlife use a range of management actions to reduce the risk of crocodile attack. Some actions are more resource intensive than others and it can take time to deploy trained staff to more remote locations. Efforts to reduce the risk posed by crocodiles include both proactive and reactive management actions.

The primary proactive management action the NT Government employs is the Be Crocwise Program. Be Crocwise engages with the public through a range of mediums including school and community presentations, merchandise and social media messaging to promote public safety and reduce risky behaviour.



PARKS AND WILDLIFE USES PROACTIVE AND REACTIVE MANAGEMENT ACTIONS REDUCING THE RISK OF CROCODILE ATTACK

SOME ACTIONS ARE MORE RESOURCE INTENSIVE THAN OTHERS.

Other management actions employed are:

- Crocodile population surveys – annual surveys to indicate population growth and movements.
- Monitoring of indicator floats –foam floats with bite marks can be an indicator of the presence of saltwater crocodiles
- Restricting access to waterways through signs, barriers and enforcement
- Crocodile surveys (day and night) – visual surveys conducted during weekly patrols and scheduled night surveys to determine crocodile numbers in high risk areas
- Harpoon capture and removal – generally at the end of the wet season or as required when problem animals are identified
- Permanently placed traps – the annual crocodile trapping program involves the placement of permanent and seasonal crocodile traps in active and barrier removal zones
- Mobile trapping – mobile traps are placed when crocodiles are sighted in a high risk area or a problem crocodile is reported
- Shooting – on occasion in high risk locations when a problem crocodile cannot be trapped it may be humanely shot either from the ground or air.

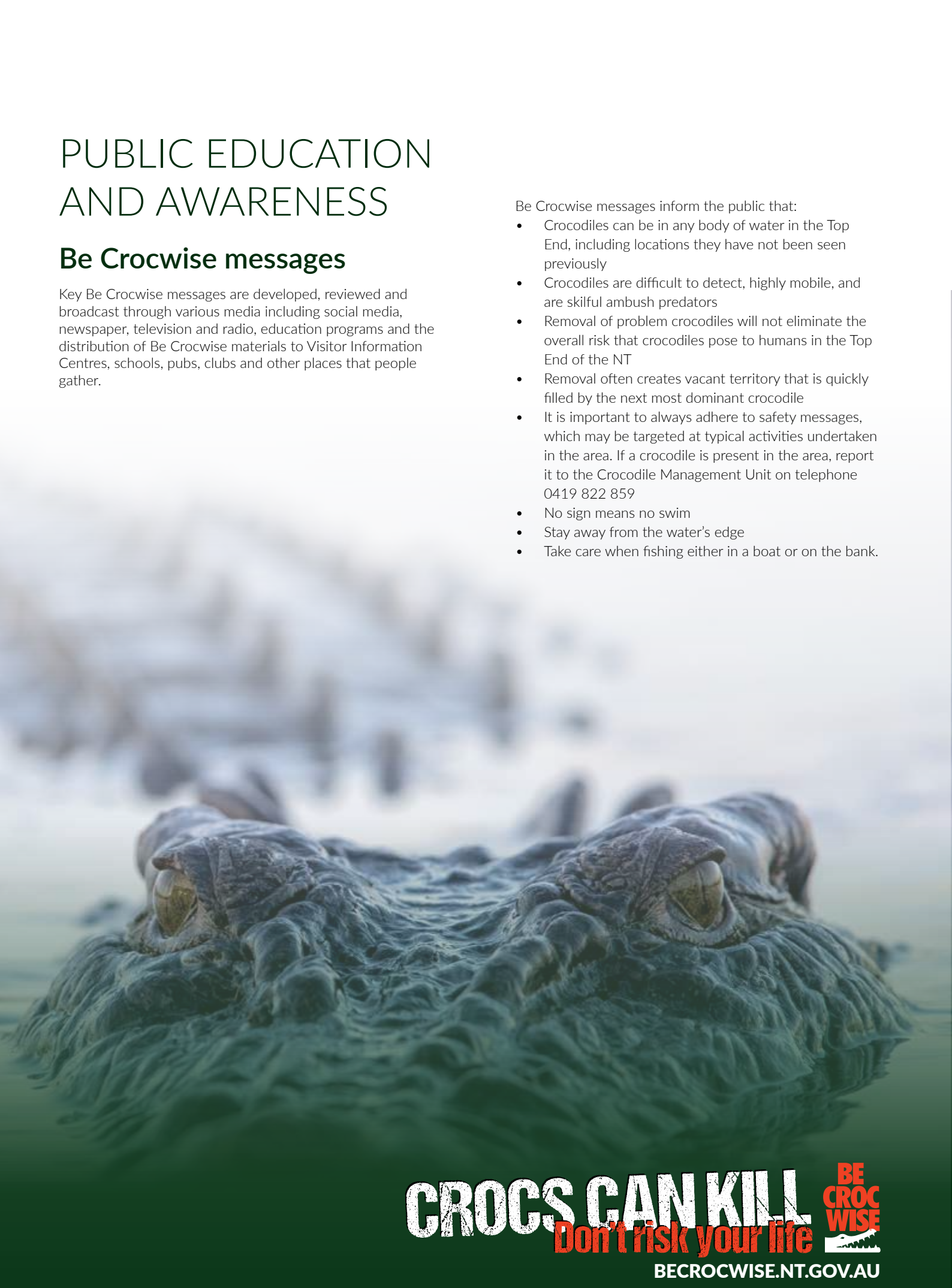
PUBLIC EDUCATION AND AWARENESS

Be Crocwise messages

Key Be Crocwise messages are developed, reviewed and broadcast through various media including social media, newspaper, television and radio, education programs and the distribution of Be Crocwise materials to Visitor Information Centres, schools, pubs, clubs and other places that people gather.

Be Crocwise messages inform the public that:

- Crocodiles can be in any body of water in the Top End, including locations they have not been seen previously
- Crocodiles are difficult to detect, highly mobile, and are skilful ambush predators
- Removal of problem crocodiles will not eliminate the overall risk that crocodiles pose to humans in the Top End of the NT
- Removal often creates vacant territory that is quickly filled by the next most dominant crocodile
- It is important to always adhere to safety messages, which may be targeted at typical activities undertaken in the area. If a crocodile is present in the area, report it to the Crocodile Management Unit on telephone 0419 822 859
- No sign means no swim
- Stay away from the water's edge
- Take care when fishing either in a boat or on the bank.



Stakeholder Management Actions

Local government councils within the Active Removal Zone include:

- City of Darwin
- City of Palmerston
- Litchfield Council
- Coomalie Community Government Council
- Katherine Town Council
- East Arnhem Regional Council
- Roper Gulf Regional Council
- Victoria Daly Regional Council.

Local governments will forward reports of potential problem crocodiles to the Crocodile Management Unit. Parks and Wildlife work closely with the councils and when there is a confirmed sighting they collaborate through the removal process.

Parks and Wildlife also meet with community organisations and assist in the development and implementation of procedures and protocols to respond to the risk before and during events held in or near the water. Community organisations such as the Darwin Surf Life Saving Club Northern Territory and the Darwin Sailing Club have a 'Crocodile Hazard Procedure' that outlines the controls to minimise the risk of crocodile attack during events and to report crocodile sightings.



Saltwater Crocodile Safety Signage

Staff will review, update and install signage according to the Parks and Wildlife's 2020 Crocodile Safety Signage Guidelines on land owned or managed by Parks and Wildlife. To obtain a copy of these Guidelines please contact the Interpretation Unit on telephone 8999 3478.

Local governments are responsible for signage elsewhere with the key messaging of 'no sign means no swimming'. Parks and Wildlife will assist local governments and others with templates for signs and advice regarding placement.



PLAN CURRENCY AND APPROVALS

This Plan will be reviewed every five years or if there is a significant incident that requires a review of the plan. A revised plan will integrate any learnings and population changes that have occurred in the interim.





Appendix A. History of Crocodile Attacks/Incidents

General Management Zone

Prior to Parks and Wildlife keeping accurate records of attacks, there have been various fatalities and injuries across the NT.

Since 1971: There has been 23 fatalities and 51 injuries from saltwater crocodile attacks in the NT which is the area encompassed by the General Management Zone. The last fatality occurred in October 2018 at a billabong at Gan Gan in East Arnhem Land. The most recent attack occurred in April 2020 at Ngangalala community on the Glyde River in Arnhem Land which resulted in an injury.

Active Removal Zones

Since 1971: Darwin Active Removal Zone - there has been 1 reported fatality and 5 injuries from saltwater crocodile attack in the area covered by this zone. The fatality occurred near the Black Jungle Conservation Reserve in March 2009 and the last injury occurred in May 2015 to a kite surfer 100 metres off the Lee Point beach.

Katherine Active Removal Zone – there has been no reported fatalities or injuries from saltwater crocodile attack in this specific zone.

Borroloola Active Removal Zone – there has been 1 reported fatality and injury from saltwater crocodile attack in the area covered by this zone. The fatality occurred in September 1986 when the person was swimming. The injury occurred in May 1985 to a person also swimming.

Nhulunbuy Active Removal Zone - there has been no reported fatalities or injuries from saltwater crocodile attacks in this specific zone.

Barrier and Removal Zones

Since 1971: There has been 1 reported injury and no fatalities from saltwater crocodile attacks in these specific zones. The injury occurred in July 2011 at Berry Springs Nature Park to a person who approached a 1.5 metre captive escapee saltwater crocodile underwater.



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